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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

COMMON TO B.E – CSE / B.TECH – IT / AIDS & M.TECH - CSE

20IT204 - COMPUTER ORGANIZATION AND ARCHITECTURE

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO1** **The addressing modes involved in the following instruction are MOV EAX, 0x10.**
   a) Register, Register b) Register, Memory
   c) Register, Immediate d) Immediate, Memory

2. **CO3** **Which of the following is virtual memory?**
   a) RAM b) Cache c) Hard Disk d) ROM

3. **CO3** **The method which offers higher speeds of I/O transfers is ___________.**
   a) Interrupts b) Memory mapping c) Program-controlled I/O d) DMA

4. **CO2** **The main advantage of multiple bus organization over single bus is _______.**
   a) Reduction in the number of cycles for execution
   b) Increase in size of the registers
   c) Better Connectivity d) Reduce memory size

5. **CO4** **A microprogram written as string of 0's and 1's is a __________.**
   a) Symbolic microinstruction b) binary microinstruction
   c) symbolic program d) binary micro-program

6. **CO5** **Structural hazards happen due to ___________.**
   a) too many instructions in pipeline
   b) un-executable instructions in pipeline
   c) enough execution units not being available
   d) Error in an execution unit

7. **CO6** **Which algorithm followed in most of the systems to perform out of order execution?**

a) Tomasulo's algorithm b) Banker's Algorithm

c) Dijkstra Algorithm d) Moor's Algorithm

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO1** SPEC termed as __________.
9. **CO1** _______ is the basic performance equation.
10. **CO3** ________ signal is used to show the completion of memory operation.

PART - B **(5x 3 = 15 Marks)**

11. **CO1** Compare RISC vs CISC.
12. **CO2** Consider the following expression

    X = (A+B) * (C+D)

    Evaluate the above expression using two address instructions.
13. **CO3** What do you mean by hit ratio?
14. **CO4** Compare horizontal and vertical micro instructions.
15. **CO1** Define GPU and write any 2 next generation processors.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

16. **CO2** i) Apply Booth's multiplication algorithm for multiplying +13 and -6. (7)

    CO2 ii) Using the restore division algorithm solve the problem. Dividend =20, Divisor = 3. (8)

17. **CO1** i) What are addressing modes? Explain various addressing modes with example. (8)

    CO1 ii) Explain the different types of functional units in a computer. (7)

18.	**CO3**	i)	Derive an execution control sequence for the instruction ADD (R3), R1 using a single bus CPU.	(7)
	CO2	ii)	Multiply the following pair of singed 2's complements numbers using bit-pair recording of the multiplier. Multiplicand= +23, Multiplier = -20.	(8)
19.	**CO3**	i)	Consider a cache consisting of 128 blocks of 16 words each, for a total of 2048 words (2K words) and assume that the main memory is addressable by 16bit address and can accommodate 4K blocks of 16 words each. Analyze various cache mapping techniques to map the main memory blocks to the cache memory for the above scenario.	(8)
	CO3	ii)	Explain the operation of DMA controller in transferring data between memory and peripherals.	(7)
20.	**CO5**	i)	Discuss in detail about data hazards and explain the technique used to overcome data hazard.	(8)
	CO3	ii)	Summarize the characteristics of different types of ROM with its usage.	(7)
21.	**CO4**	i)	Compare hardwired and micro programmed control unit.	(8)
	CO6	ii)	Illustrate about GPU and its impact on the modern processors.	(7)
22.	**CO6**	i)	What is Dynamic scheduling? Explain in detail about various steps involved in Tomasulo's Algorithm with example.	(10)
	CO6	ii)	Explain Instruction level parallel processing and state the challenges of parallel processing.	(5)

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